

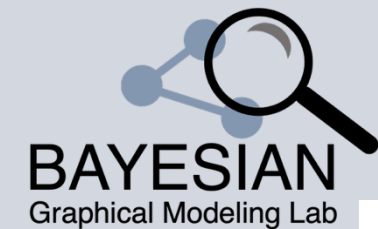
Bayesian Graphical Modeling

Topological Priors

What is psychometrics? 2025

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bayesiangraphicalmodeling.com



Session setup



Part I: Theory



Part II: Tutorial / Practical



Session setup



Part I: Theory

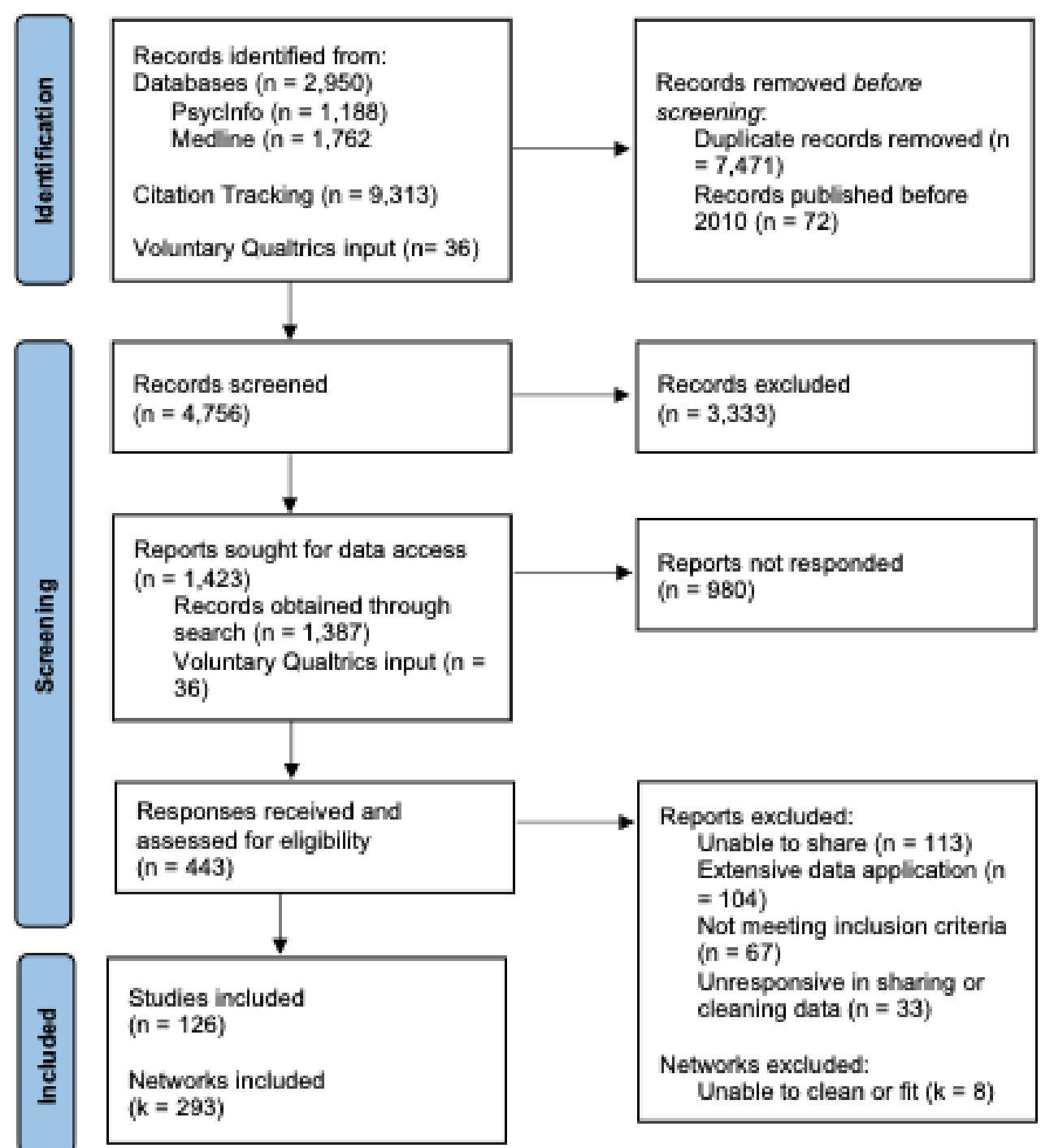
Whats the evidence?

Contrary to most existing network estimation methods, the **Bayes factor analyses** in Session I allowed us to ***quantify evidence of absence***, not just detect ***absence of evidence***.

This naturally raises a question: **What happens when we apply this Bayesian reasoning to existing, published networks?**

To explore this, we performed a **Bayesian reanalysis** of previously reported networks.

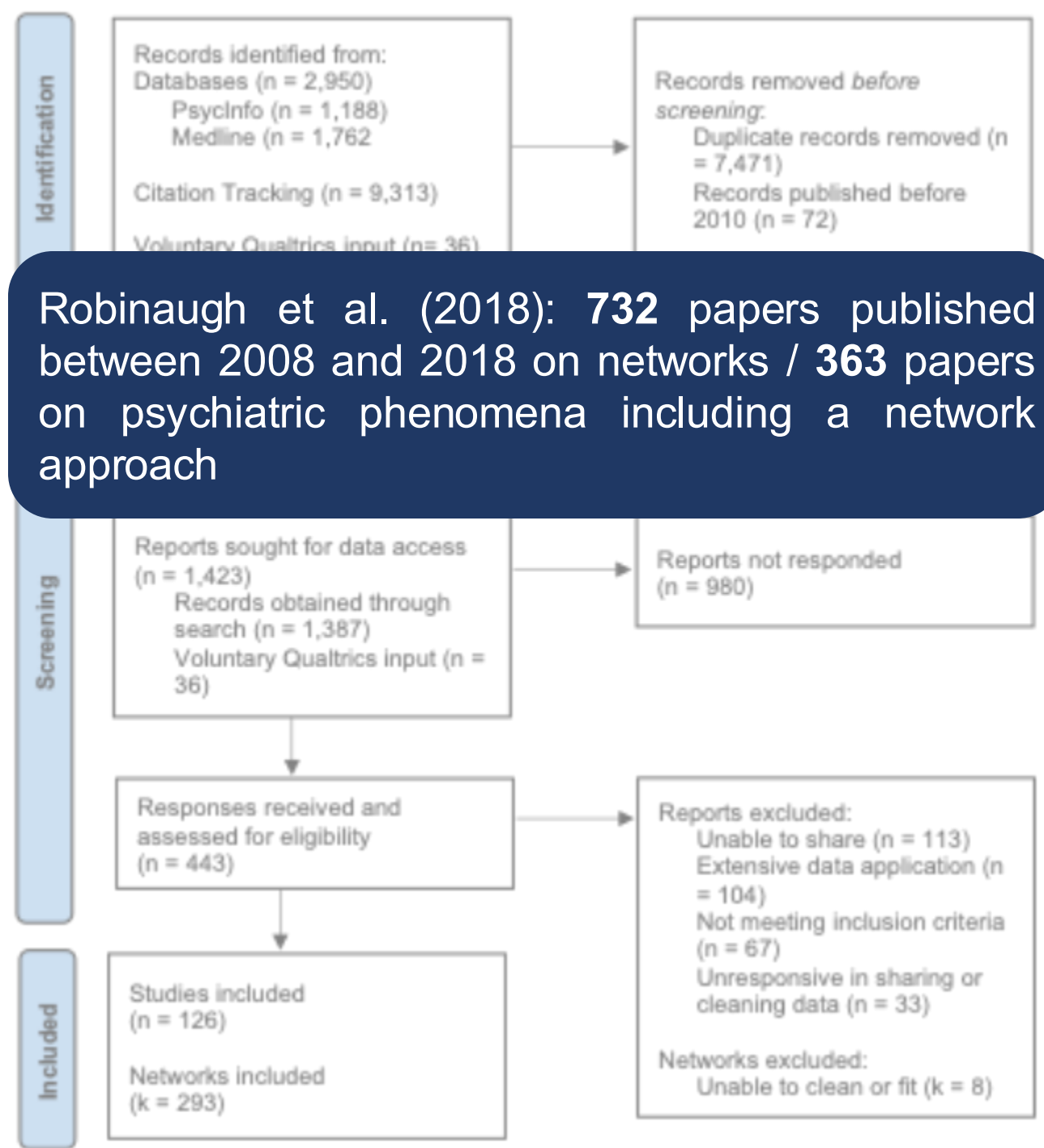
Papers and data



Papers and data

Screening 4,720 papers

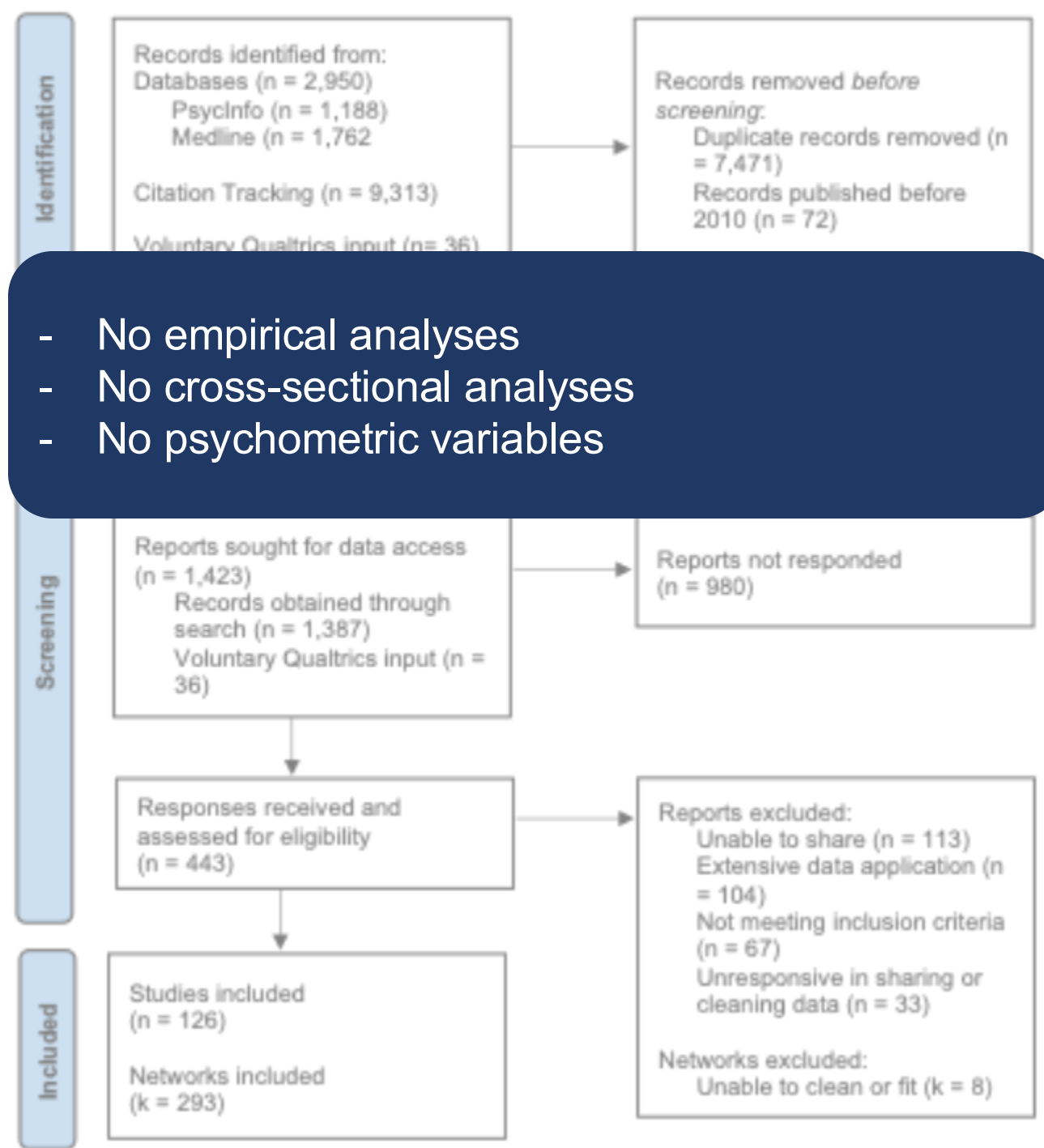
- 3,333 excluded
- 1,387 included (+ 36 additions)



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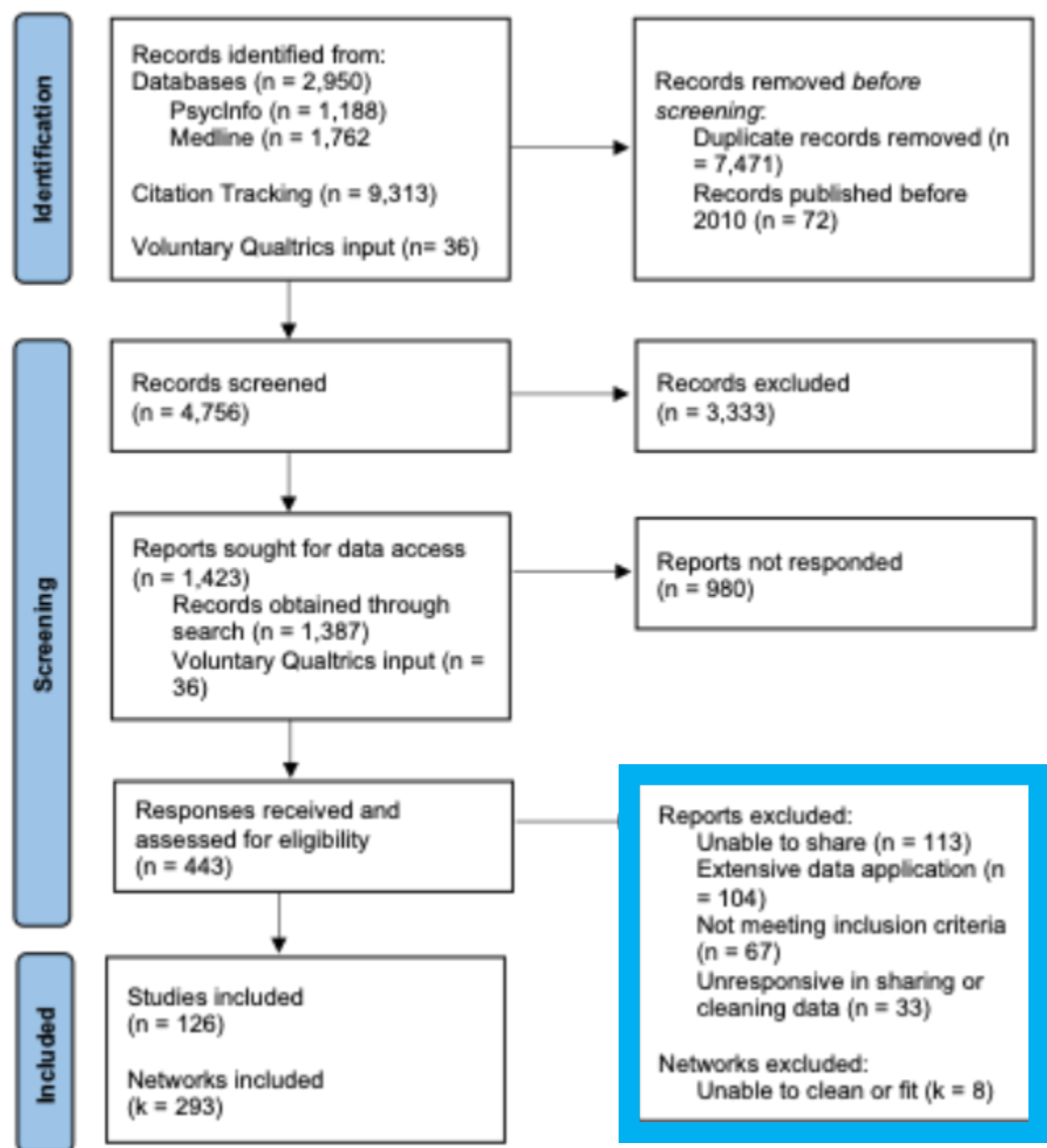
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Inquired 1,423 datasets

- 443 responses
 - 126 included
 - 317 excluded

→ 126 papers; 293 networks



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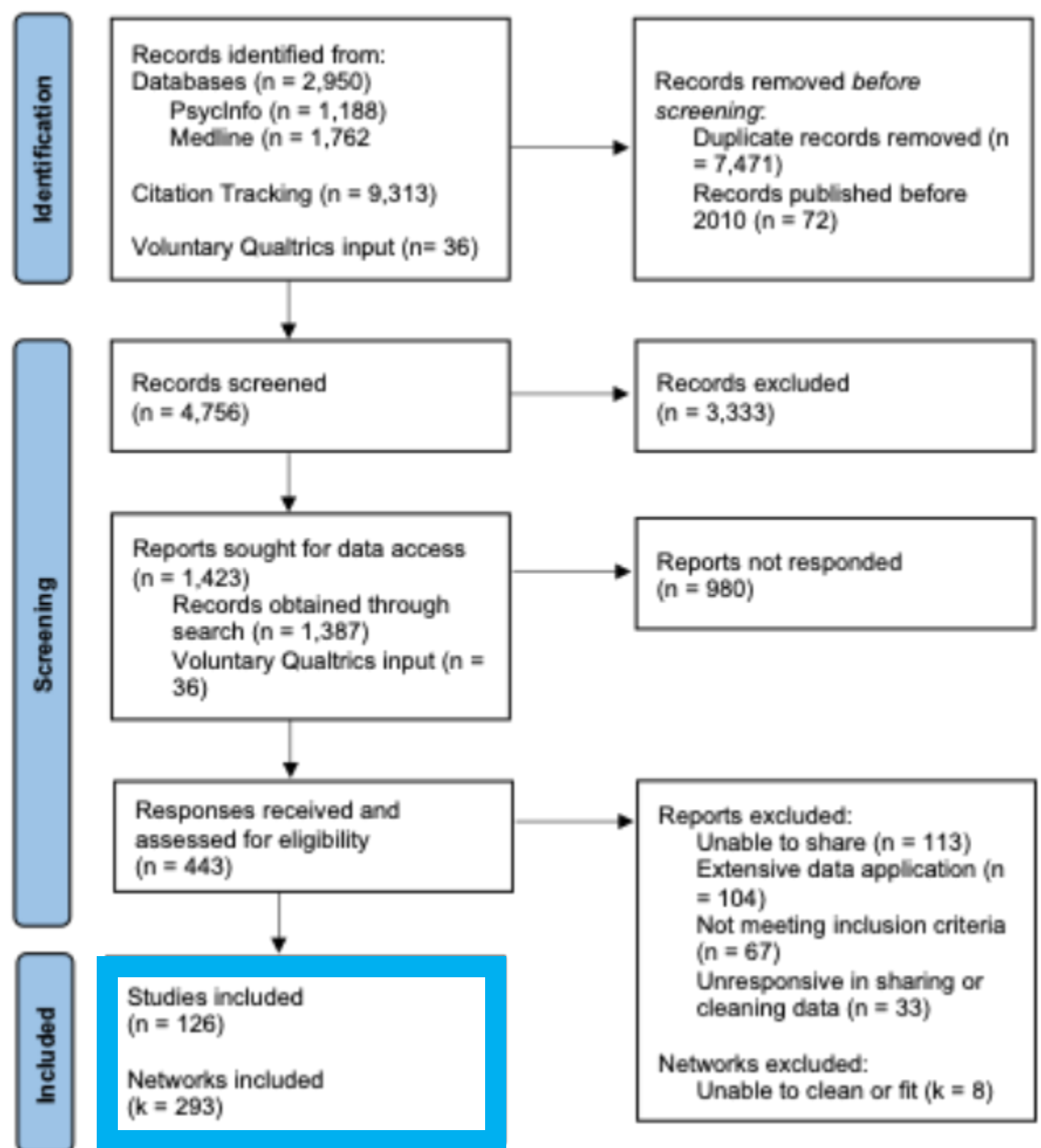
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Analysis

We analyzed all networks using the Gaussian graphical model

For each network relation, we use the Bayes factor to assess the support in the data for the competing hypotheses:

$\mathcal{H}_0$: Partial association is zero

$\mathcal{H}_1$: Partial association is non-zero

Analysis

The Bayes factor indicates how much more plausible the data are under one hypothesis relative to the other

$$\text{BF}_{10} = \frac{p(\text{data} \mid \mathcal{H}_1)}{p(\text{data} \mid \mathcal{H}_0)}$$

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Evidence for $\mathcal{H}_0$

Evidence for $\mathcal{H}_1$

$$\text{BF}_{10} = \frac{1}{30}$$

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$$\text{BF}_{10} = \frac{1}{3}$$

$$\text{BF}_{10} = 1$$

$$\text{BF}_{10} = 3$$

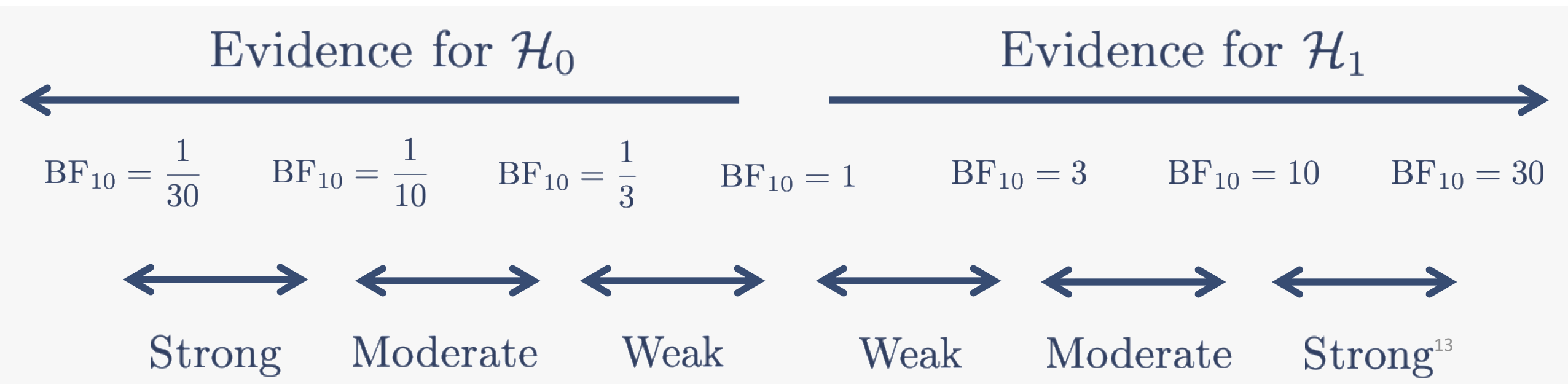
$$\text{BF}_{10} = 10$$

$$\text{BF}_{10} = 30$$

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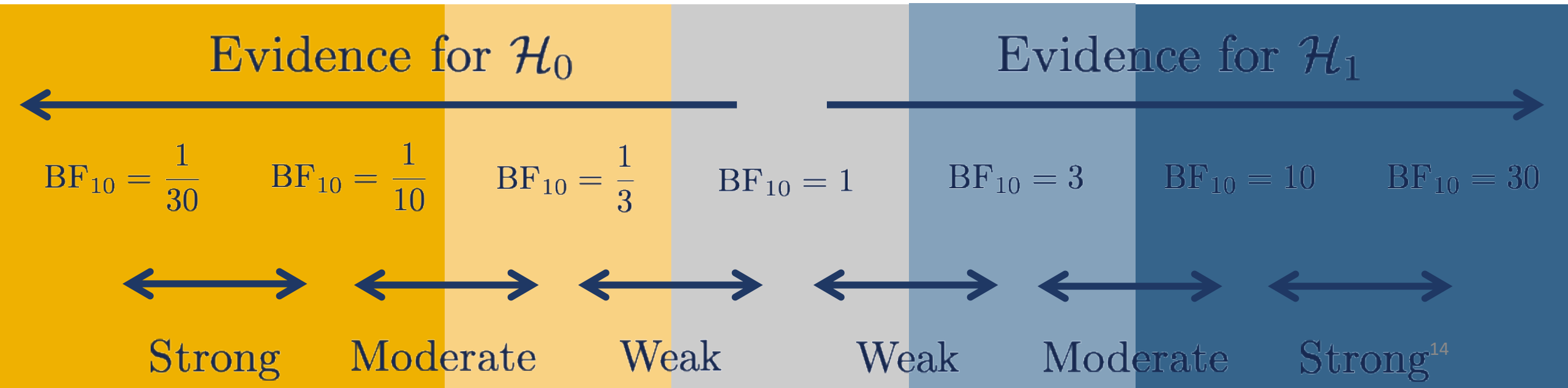
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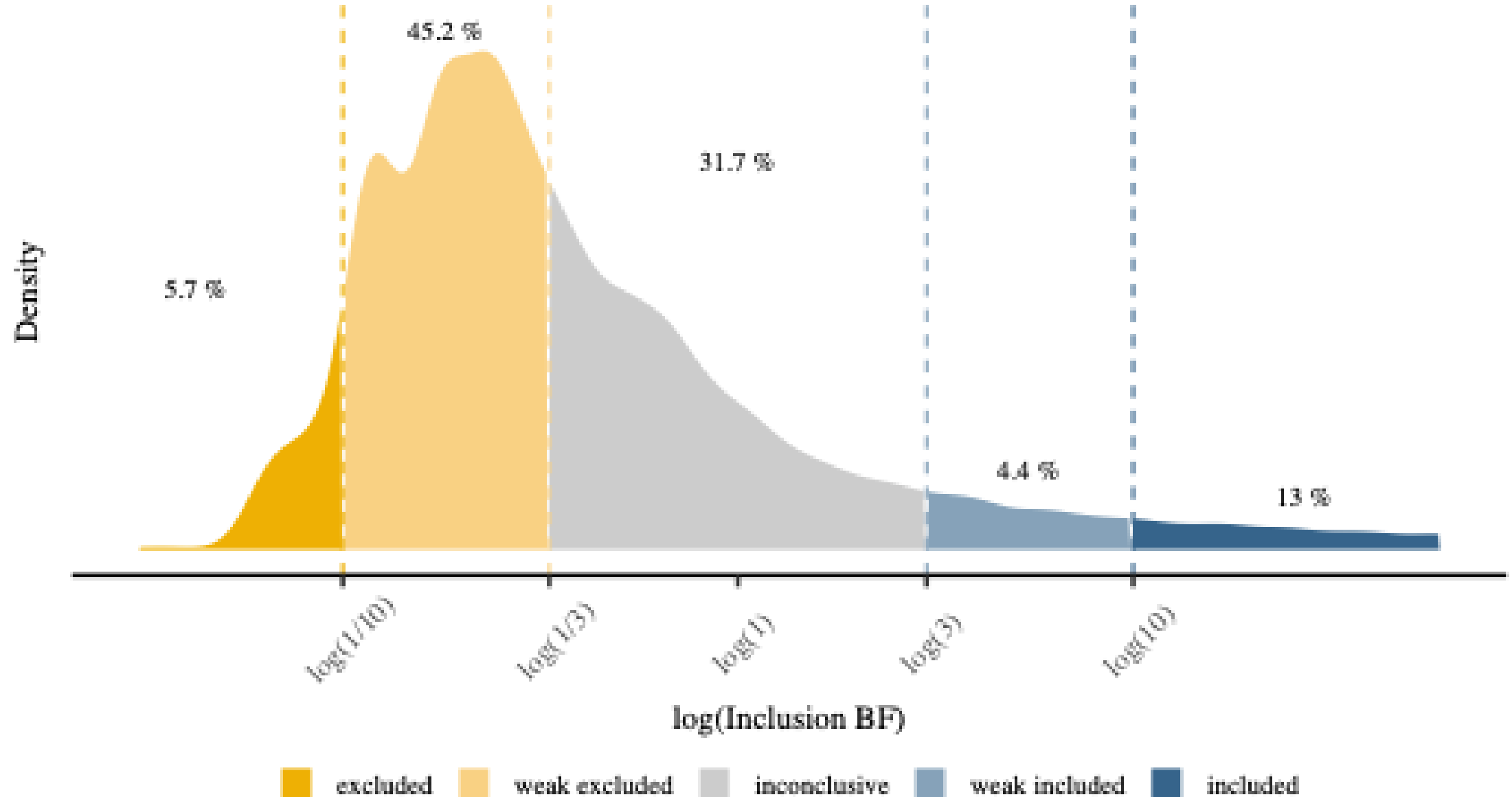
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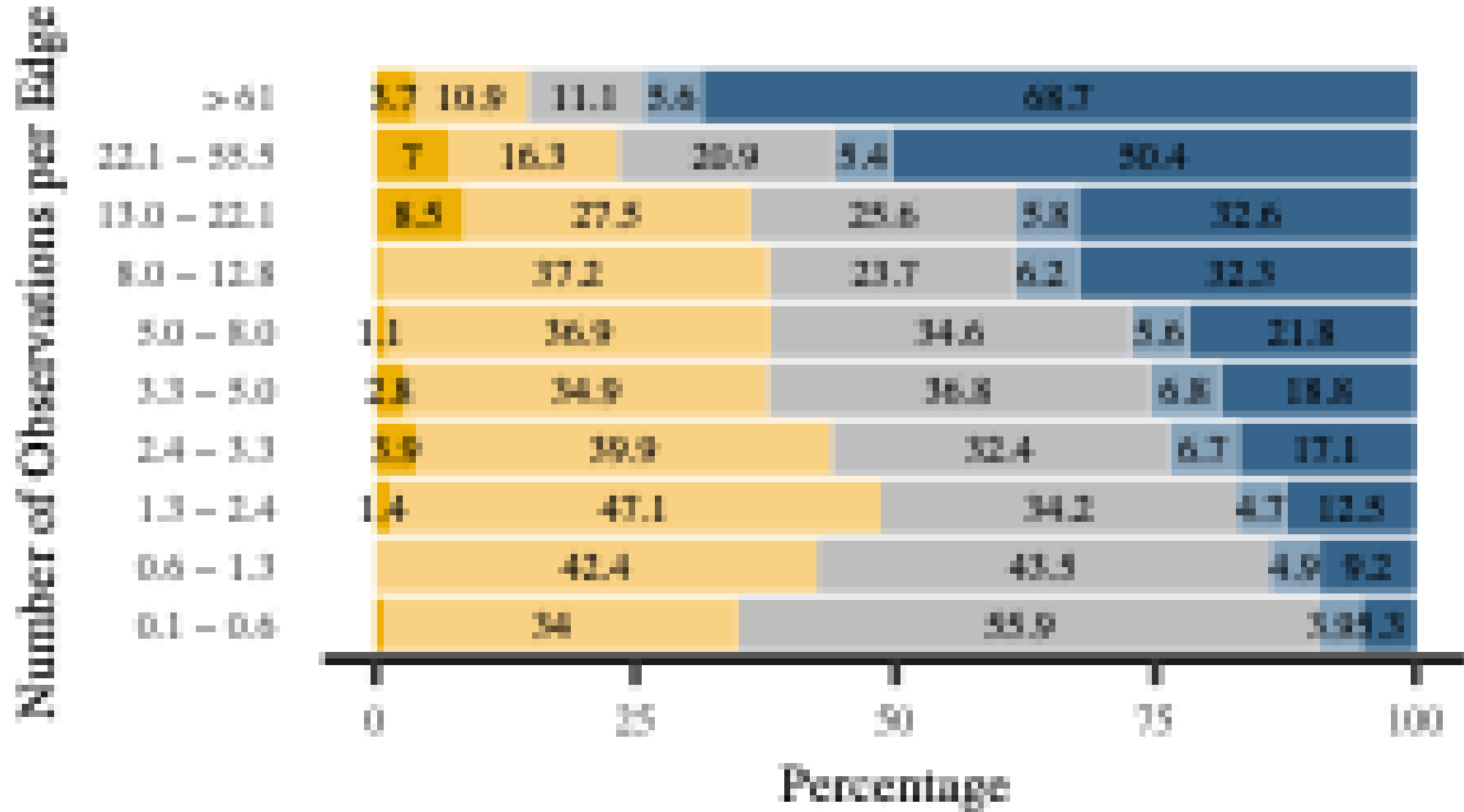
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Results

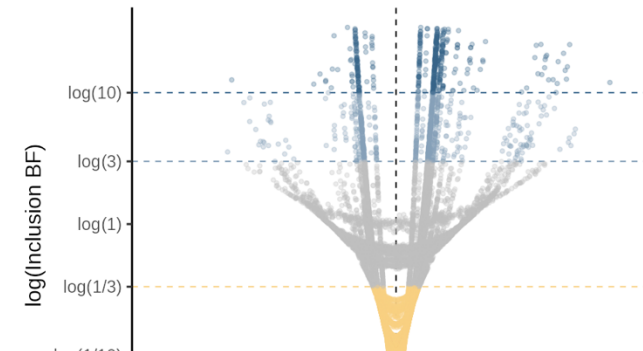
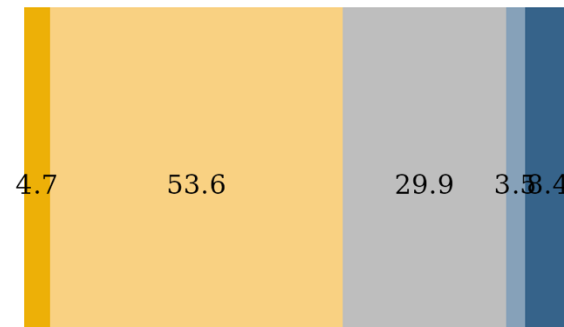
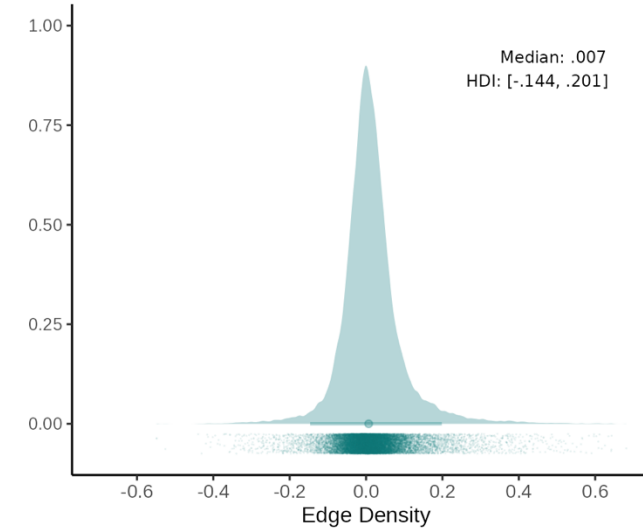
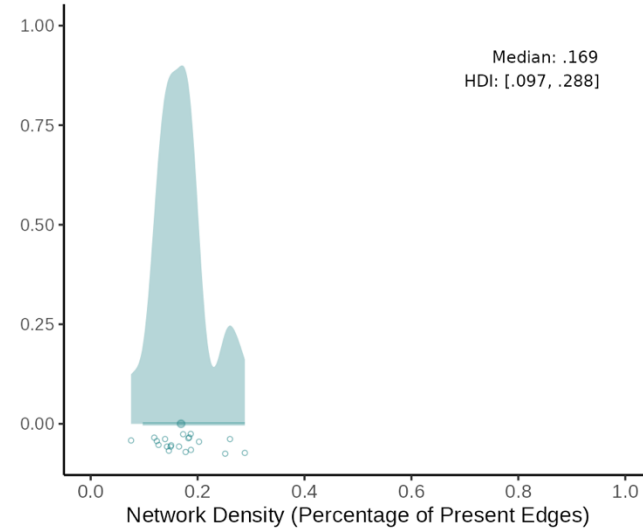
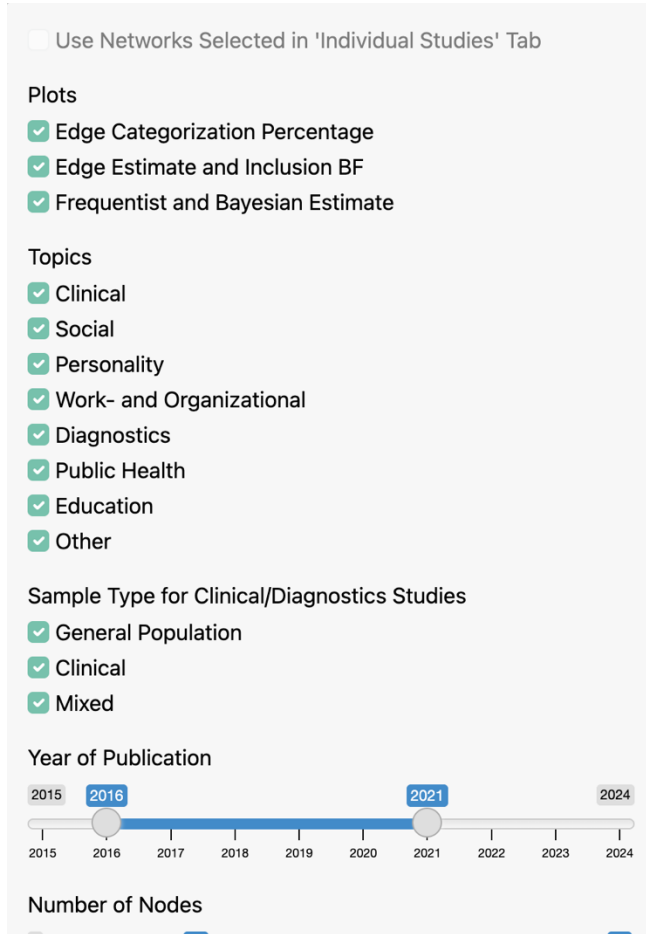


Results



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<https://uvasobe.shinyapps.io/ReBayesed/>



Conclusions

1. Strong evidence for the presence or absence of ~20% edges
 $BF_{10} < 1/10$ or $10 < BF_{10}$

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Conclusions

1. Strong evidence for the presence or absence of ~20% edges
 $BF_{10} < 1/10$ or $10 < BF_{10}$
2. Weak evidence for the presence or absence of ~50% edges
 $1/10 < BF_{10} < 1/3$ or $3 < BF_{10} < 10$
3. Inconclusive evidence of remaining ~30% of edges
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4. Strength of evidence strongly determined by (relative) sample size
Median was 5 obs. per edge ($1 < , > 7,000$).

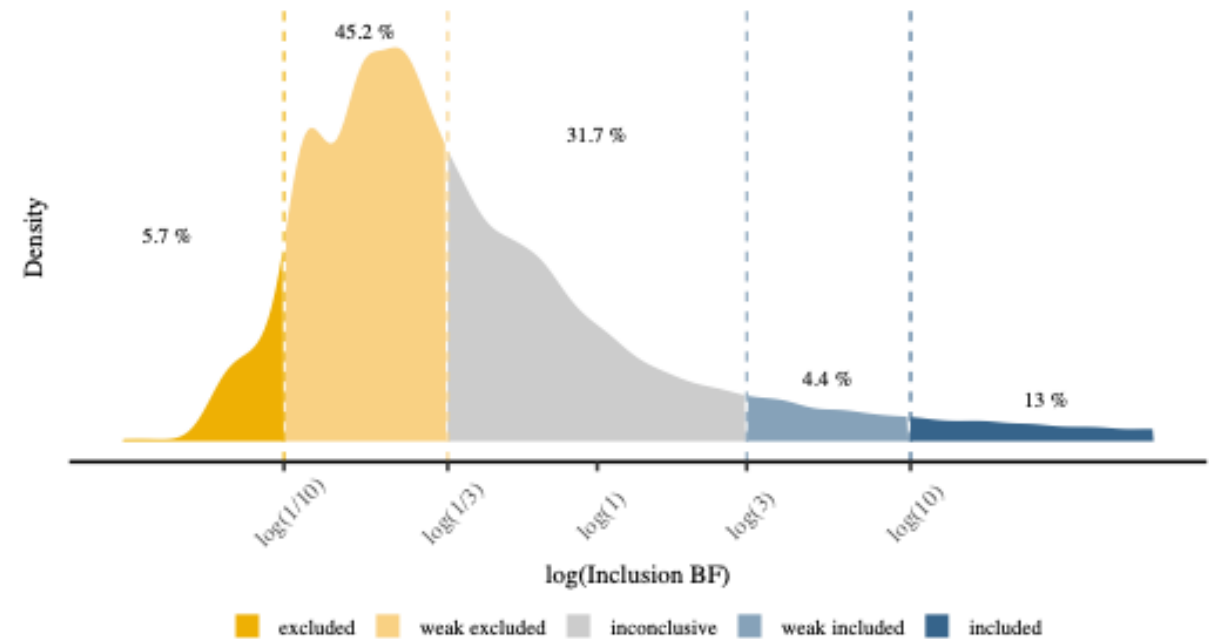
Limitations

We applied a **single-model Bayes Factor** and GGM to all data

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We did not compare to original conclusions

- Might have indicated uncertainty

- Might not have been interested in the structure/edges

What did the reanalysis reveal?

Bayes factors for most edges were **small** → evidence for inclusion or exclusion often **inconclusive**

As a result, there is **considerable uncertainty** about the overall network structure

Our Bayes factor methods **ignored multiplicity** → overstate confidence in individual edges

Together, this shows the need to **model structural uncertainty explicitly**

Existing models/priors for network structures

The independent Bernoulli model

$$\begin{aligned} p(\gamma_{12}, \dots, \gamma_{(p-1)p}) &= \prod_{i=1}^{p-1} \prod_{j=i+1}^p \pi^{\gamma_{ij}} (1 - \pi)^{1-\gamma_{ij}} \\ &= \pi^k (1 - \pi)^{m-k} \end{aligned}$$

Here, k is number of active edges, m total number of edges. But what is the inclusion probability?

Marsman, M., Huth, K., Waldorp, L., & Ntzoufras, I. (2022). Objective Bayesian Edge Screening and Structure Selection for Ising Networks. *Psychometrika*, 87, 47-82.

Mohammadi, R., & Wit, E. (2019). BDgraph: An R Package for Bayesian Structure Learning in Graphical Models. *Journal of Statistical Software*, 89, 1–30.

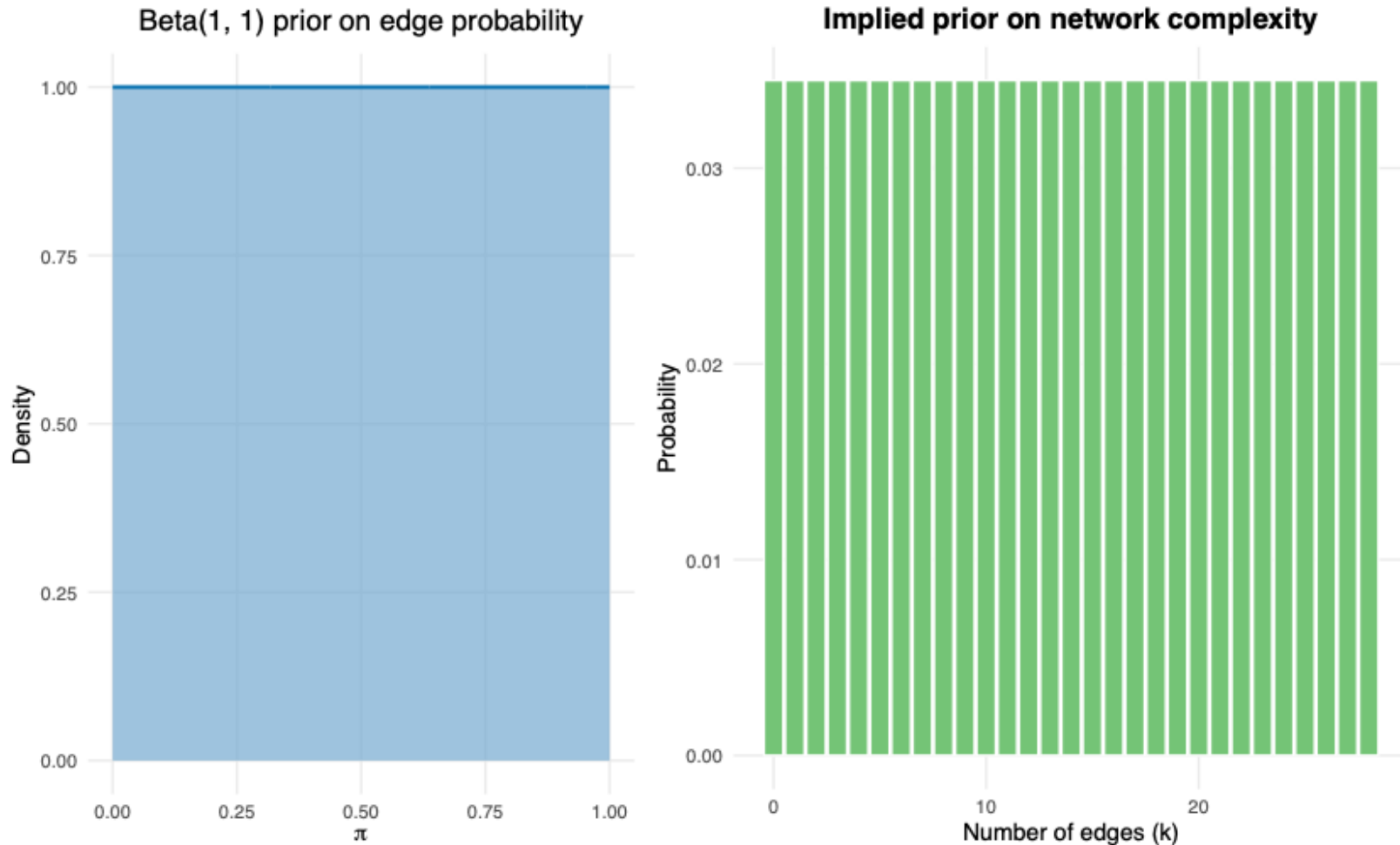
Existing models/priors for network structures

The Beta-Bernoulli model

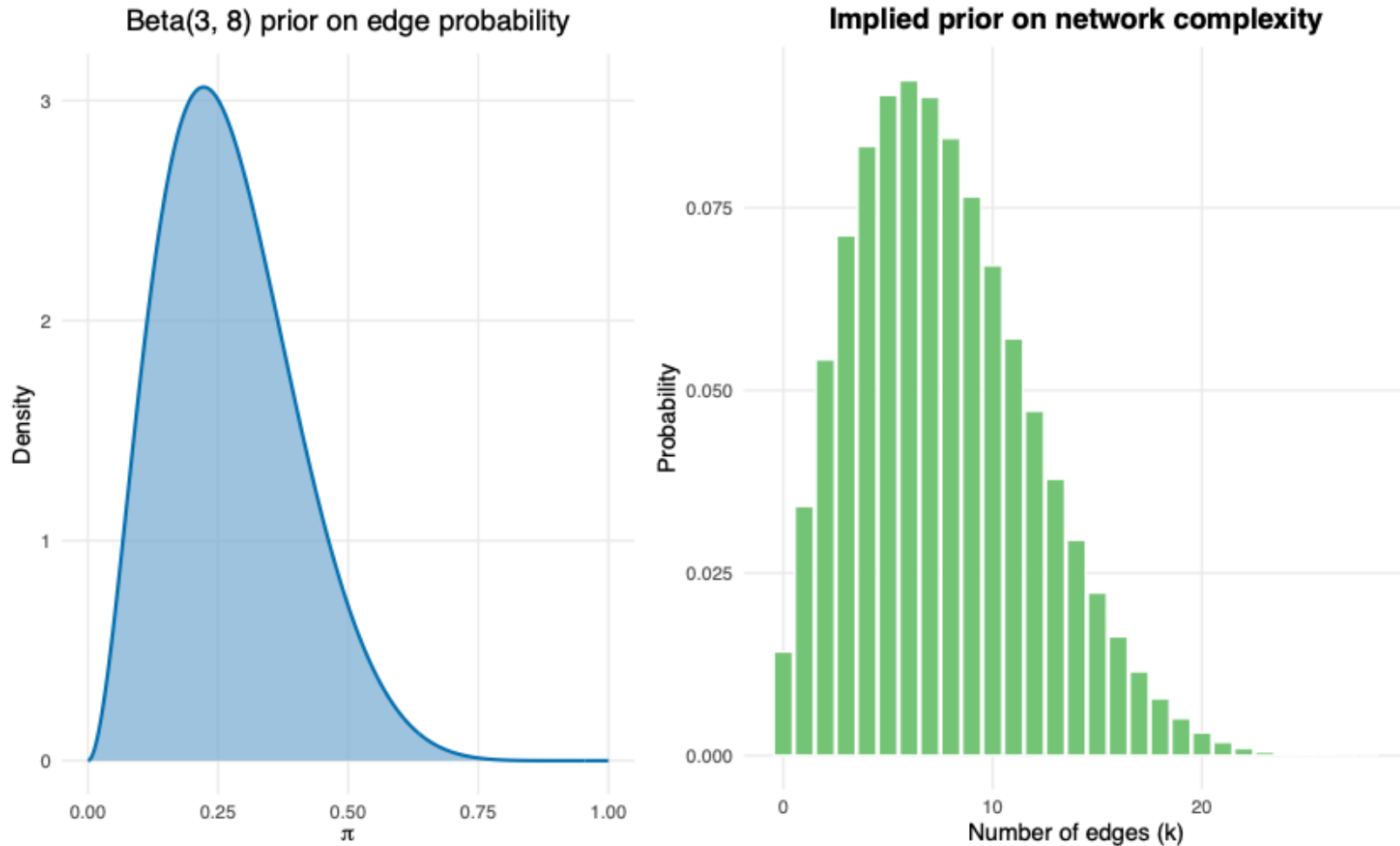
$$\begin{aligned} p(\gamma_{12}, \dots, \gamma_{(p-1)p}) &= \int_0^1 \overbrace{\prod_{i=1}^{p-1} \prod_{j=i+1}^p \pi^{\gamma_{ij}} (1 - \pi)^{1-\gamma_{ij}}}^{\text{Bernoulli model}} \overbrace{\frac{1}{B(\alpha, \beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1} d\pi}^{\text{Beta density}} \\ &= \frac{1}{B(\alpha, \beta)} B(\alpha + k, \beta + m - k) \end{aligned}$$

Here, k is number of active edges, m total number of edges.

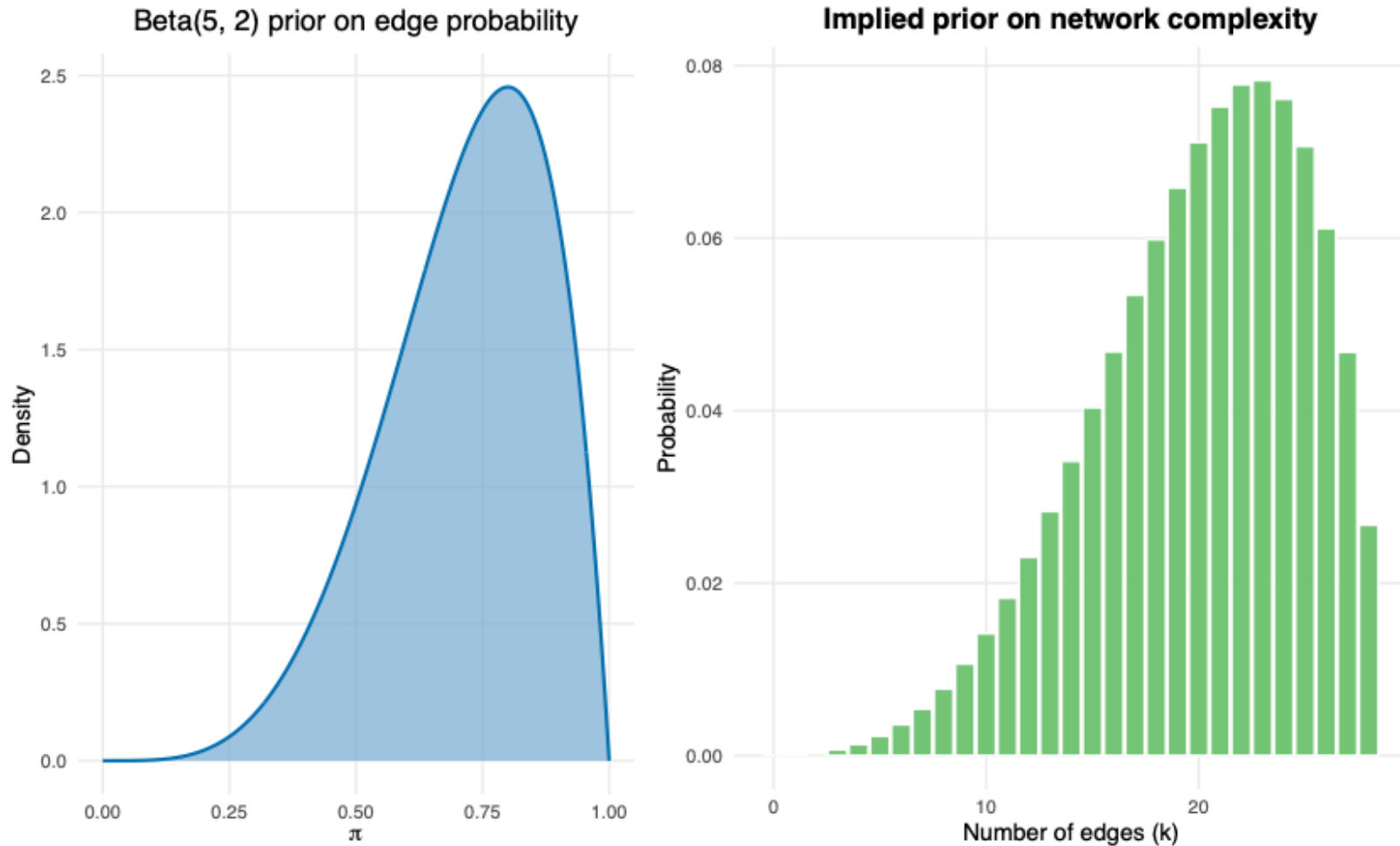
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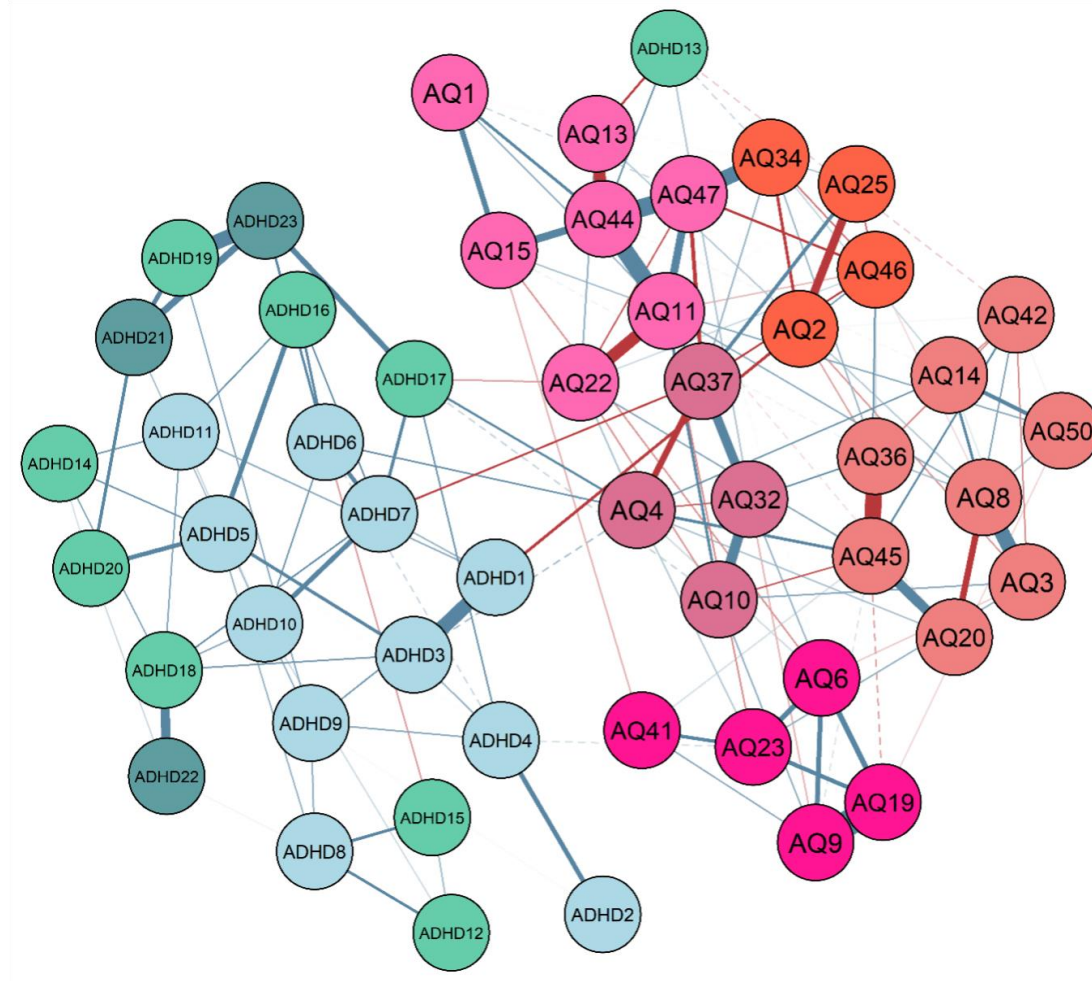
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The Stochastic-Block model

$$p(\gamma_{12}, \dots, \gamma_{(p-1)p} \mid z_1, \dots, z_p) p(z_1, \dots, z_p \mid b) p(b)$$

Here, z are cluster allocations, m total number of clusters (blocks).

Existing models/priors for network structures



From edge evidence to topological priors

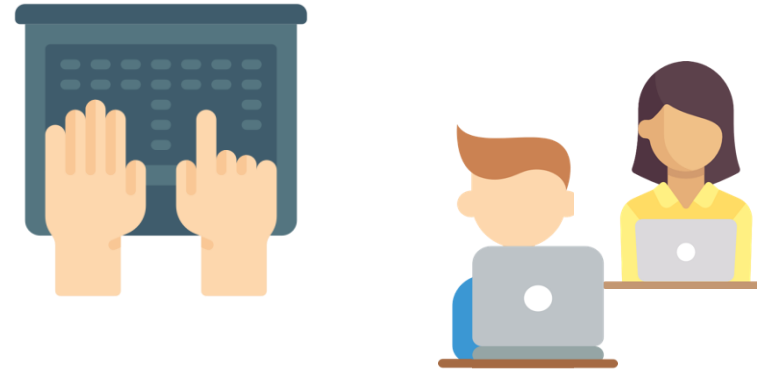
Why this matters

- Network structures show **large uncertainty**.
- Existing (Bayesian) methods **ignore multiplicity**.
- We need to **model uncertainty at the structural level**.

What we will do in the tutorial

- Learn how to **represent beliefs about topology** using priors.
- See how priors provide **automatic multiplicity correction**.
- Explore three key priors:
 - **Bernoulli** → overall density
 - **Beta-Bernoulli** → balanced complexity
 - **Stochastic Block Model (SBM)** → clustering & domains

Session setup



Part II: Tutorial / Practical